Geocomputation

Point Pattern Analysis



Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

This week

- Focus analysis directly on point events rather than aggregating to administrative geography.
- Three main ways to describe or characterise point processes:
  - Descriptive statistics.
  - Distance-based methods: e.g. average nearest neighbour.
  - Density-based methods: e.g. DBSCAN, kernel density estimation.
- Some practical examples.

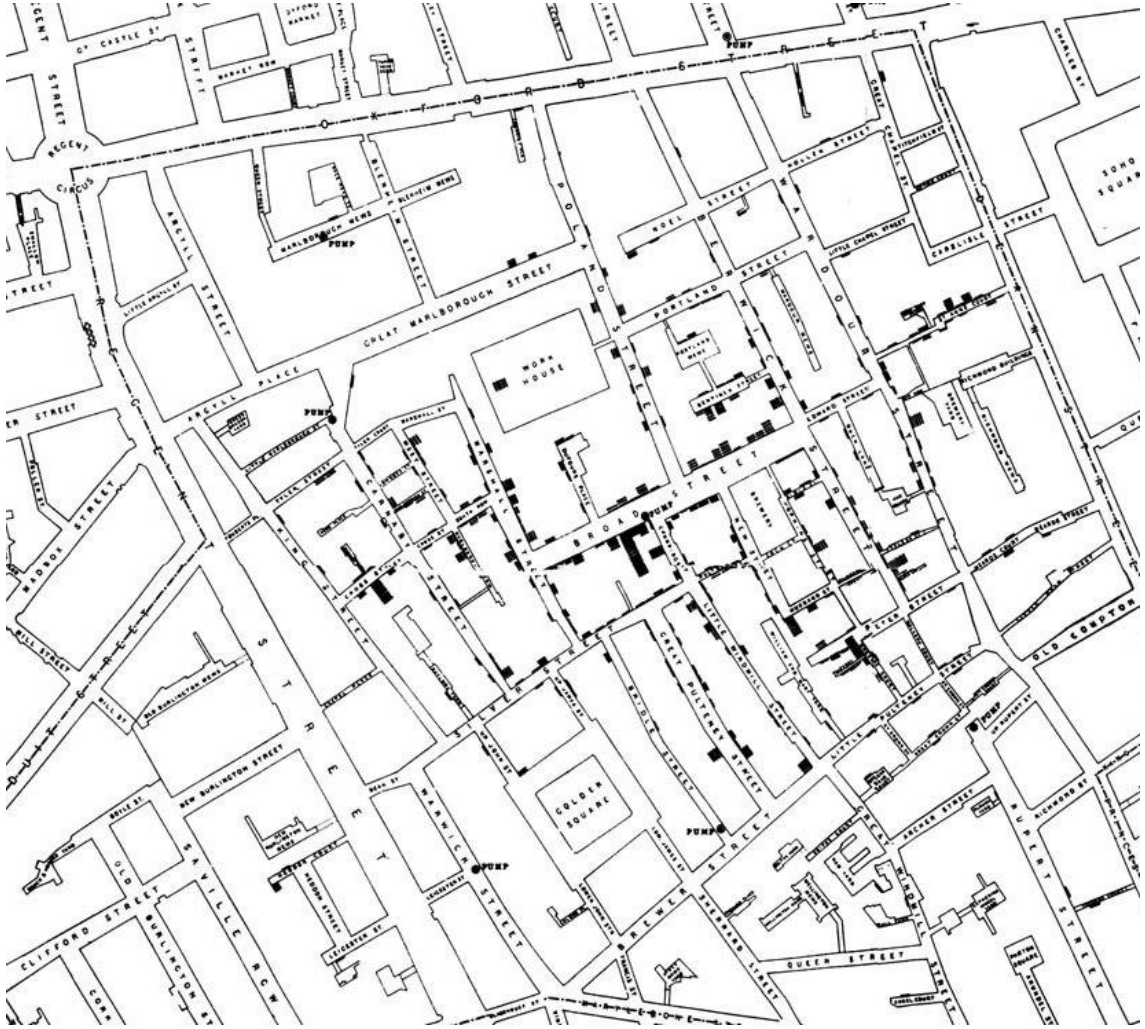
Before we start – CMD

- Go to www.menti.com
- Use code: XXXX XXXX

Why do we want to analyse points?

- Points represent the location of individuals, features, or events, providing a spatial reference for analysis.
- The set of points we observe is a point pattern; the process that produced it is a point process. Patterns in the points can reveal underlying relationships and trends.
- Points can also carry attributes (non-spatial information), such as the type of crime, so we can compare patterns between groups.

Spatial analysis

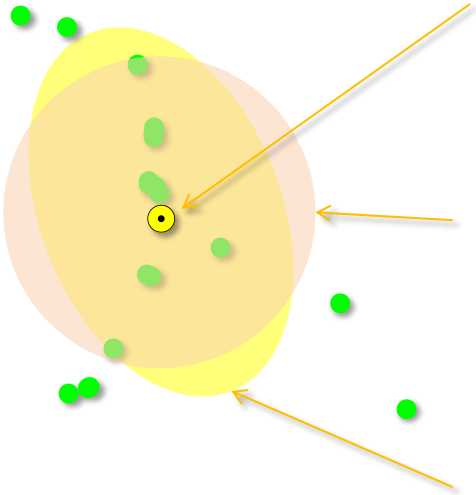


How do we analyse points?

- Descriptive statistics.
- Distance-based methods.
- Density-based methods.

Descriptive statistics

Centre and spread of points



Mean center
computed average X and Y coordinate values.

$$\bar{s} = \left(\frac{\sum_{i=1}^{n} x_i}{n}, \frac{\sum_{i=1}^{n} y_i}{n} \right)$$

Standard distance
measure of the variance between the average distance of the features to the mean center.

$$d = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \mu_x)^2 + (y_i - \mu_y)^2}{n}}$$

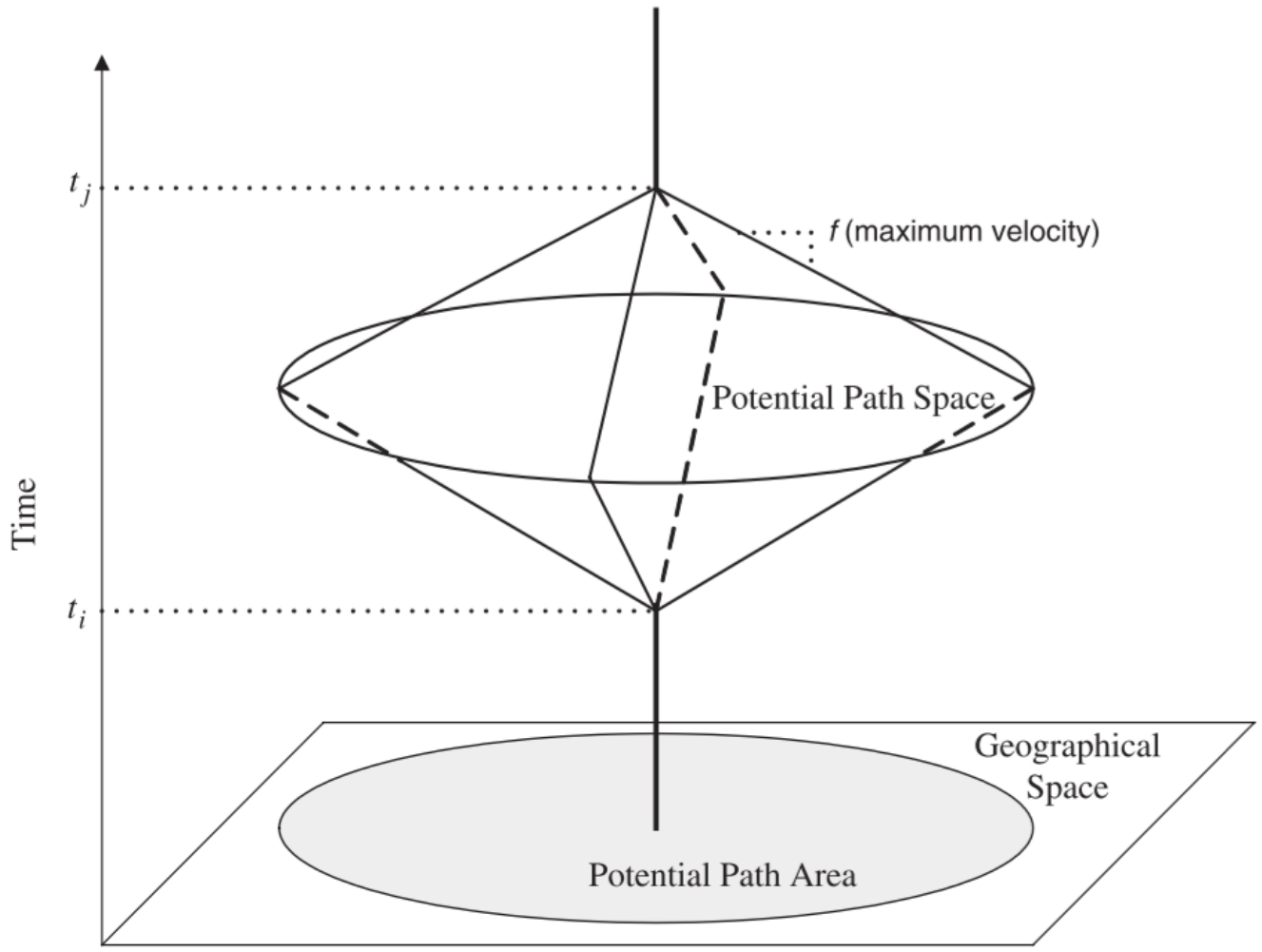
Standard deviational ellipse
separate standard distances for each axis.

$$d_x = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \mu_x)^2}{n}}$$

$$d_y = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \mu_y)^2}{n}}$$

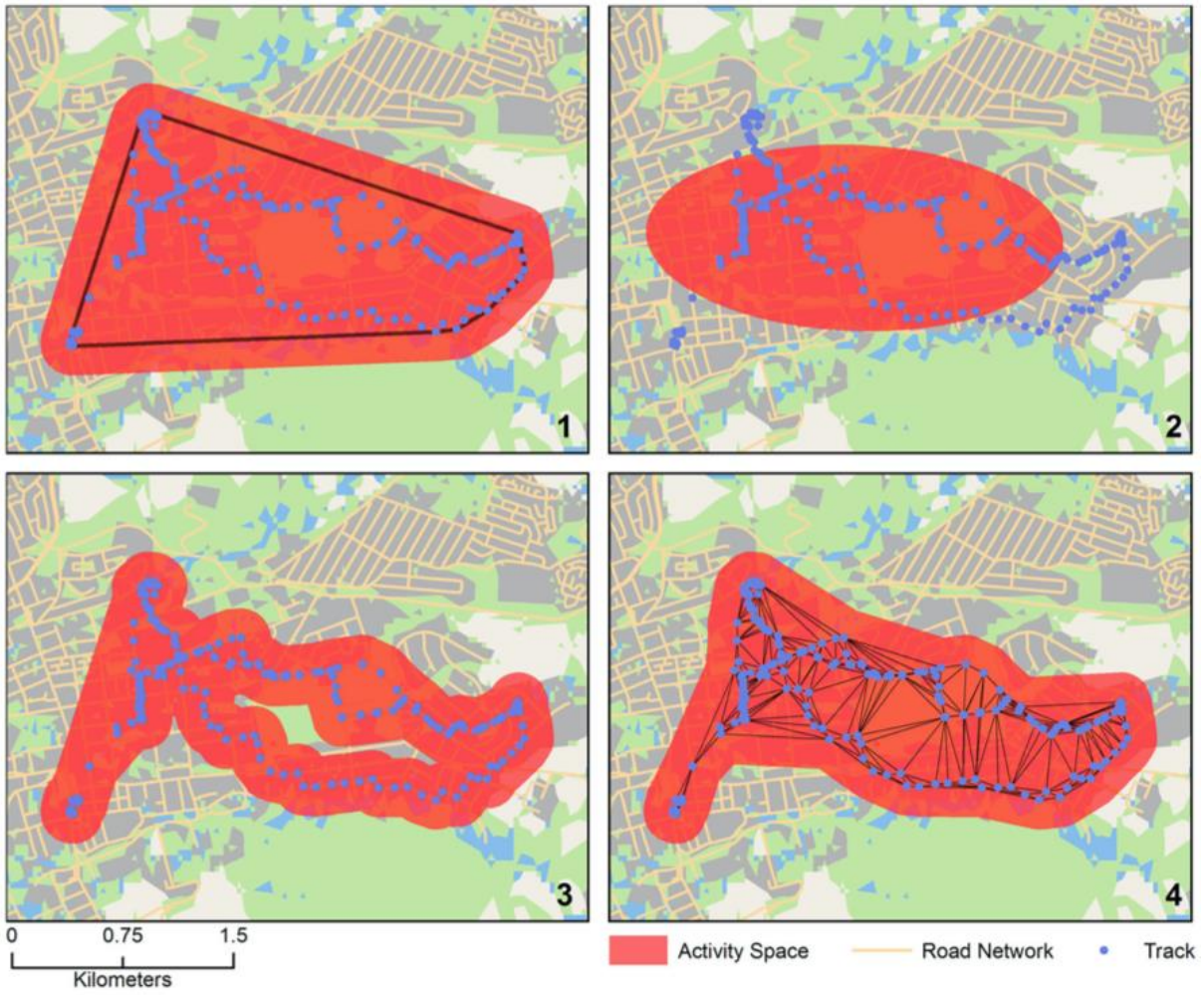
Gimond, M. 2025. Intro to GIS and Spatial Analysis. [online]
https://mgimond.github.io/Spatial/index.html

Space-time prism



Miller, 2005.

Activity space



Distance-based measures

Distance-based measures

- Using distance as a summary measure to summarise a point process.
- Measure of spread of the distribution.

Average Nearest Neighbour

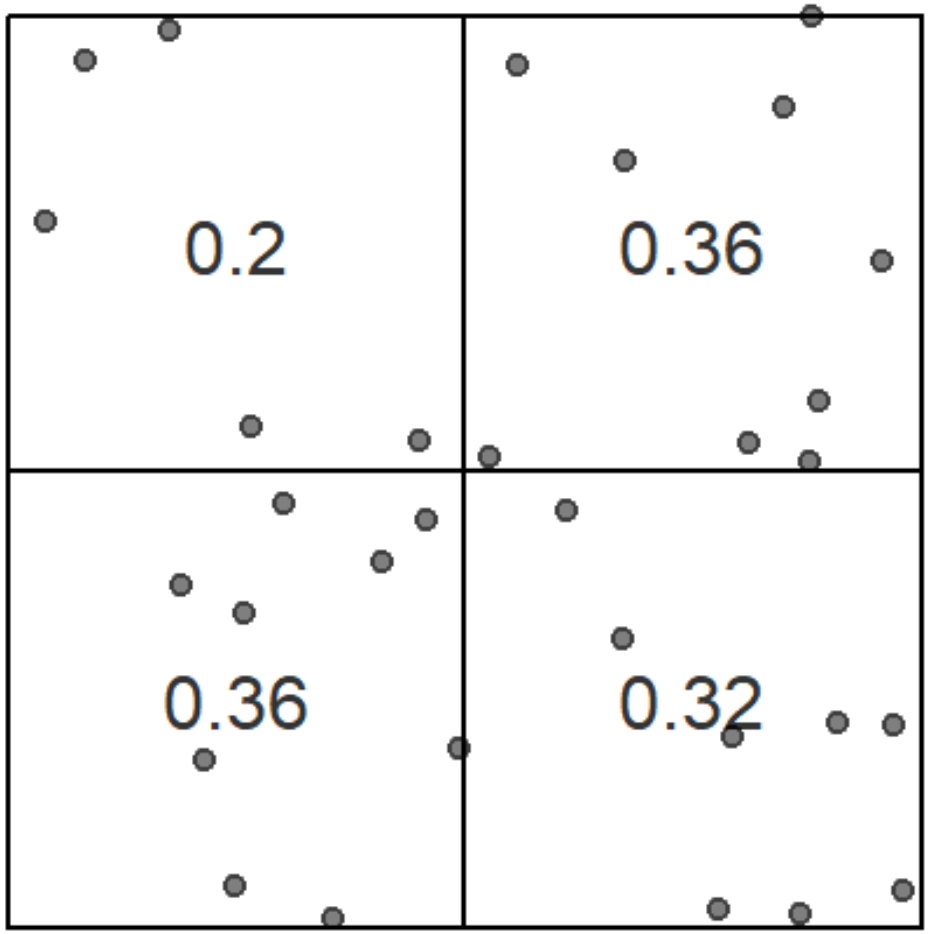
- ANN measures the distance between each feature and its nearest neighbour, then averages all these nearest-neighbour distances.
- If the average distance is smaller than the average for a hypothetical random distribution, the features are considered clustered. If the average distance is larger, the features are considered dispersed.
- Random here means that every location is equally likely to contain a point: complete spatial randomness (CSR), which we return to in week 4.

Density-based measures

Density-based measures

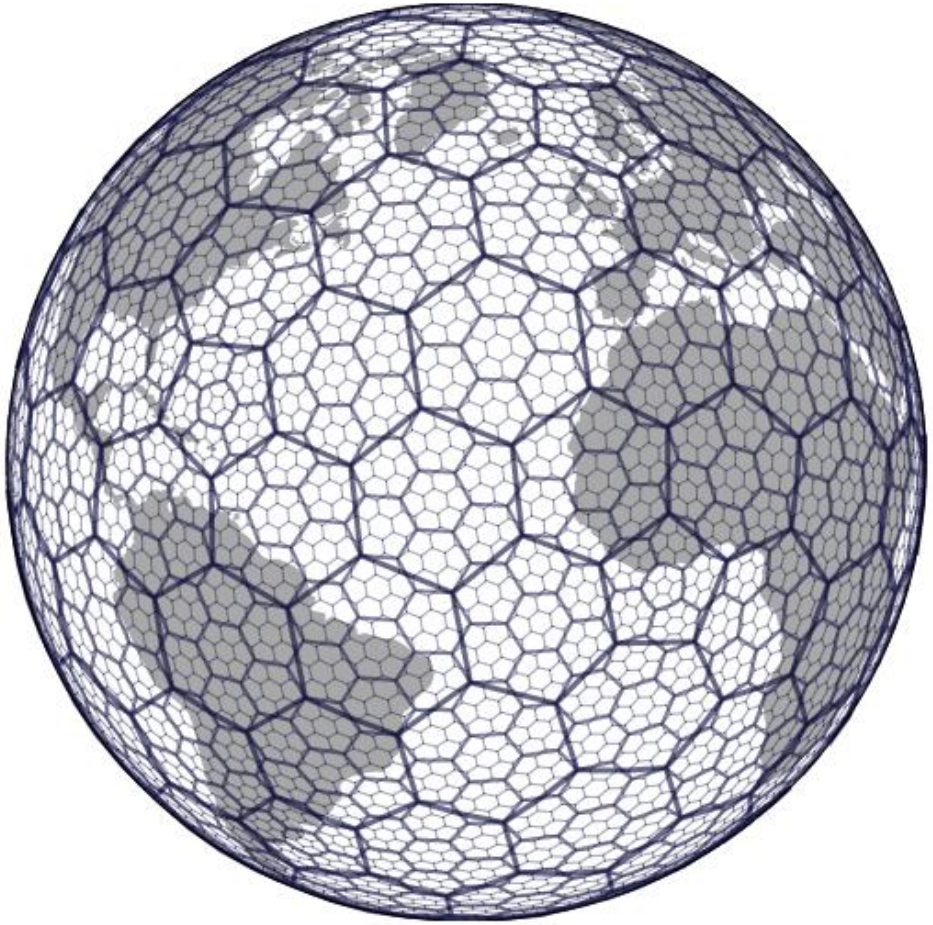
- Measure of the intensity of a point process: the number of points per unit area.
- Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to find high-density clusters and distinguish noise points.
- Kernel Density Estimation (KDE) smooths point data to estimate the density of points across a given area.

Density-based measures



Gimond, M. 2025. Intro to GIS and Spatial Analysis. [online]
https://mgimond.github.io/Spatial/index.html

Hexagonal bins instead of squares

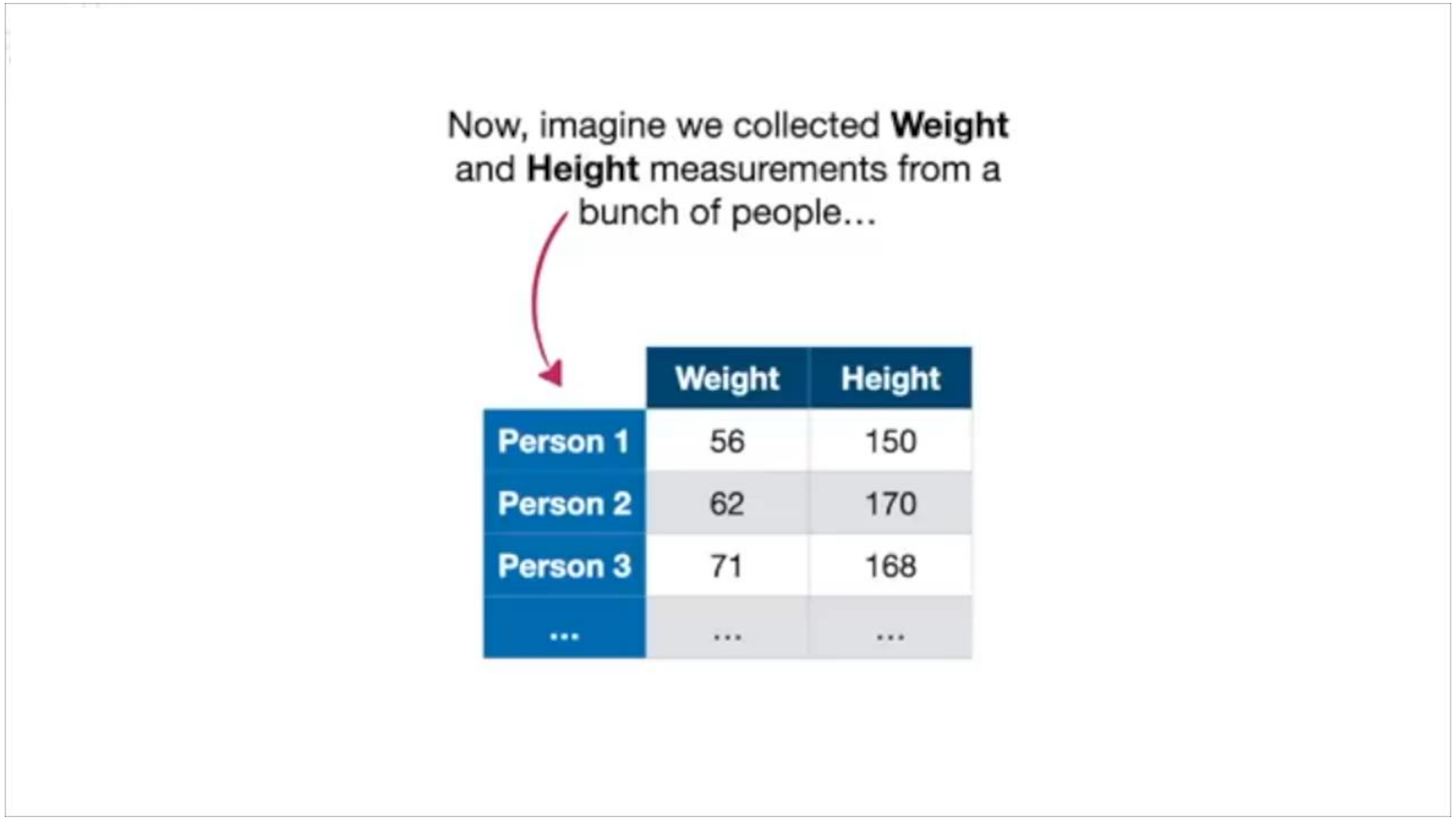


Uber. 2018. *H3: Uber's Hexagonal Hierarchical Spatial Index*. [online] https://www.uber.com/blog/h3/

DBSCAN

- Density–Based Spatial Clustering of Applications with Noise (DBSCAN).
- Functionally, DBSCAN detects clusters of points and noise by evaluating density.
- It relies on two key parameters: `minPts` (the minimum number of points within the search radius for a point to count as dense) and `eps` (epsilon, the search radius: the maximum distance between two points for them to be neighbours).
- A point with at least `minPts` points within `eps` of it (itself included) is a core point. Clusters grow from core points: points within `eps` of a core point join the cluster, and every other point is noise.

DBSCAN



	Weight	Height
Person 1	56	150
Person 2	62	170
Person 3	71	168
...	...	...

StatQuest. *Clustering with DBSCAN*. [online]
https://www.youtube.com/watch?v=RDZUdRSDOok

DBSCAN

Pros:

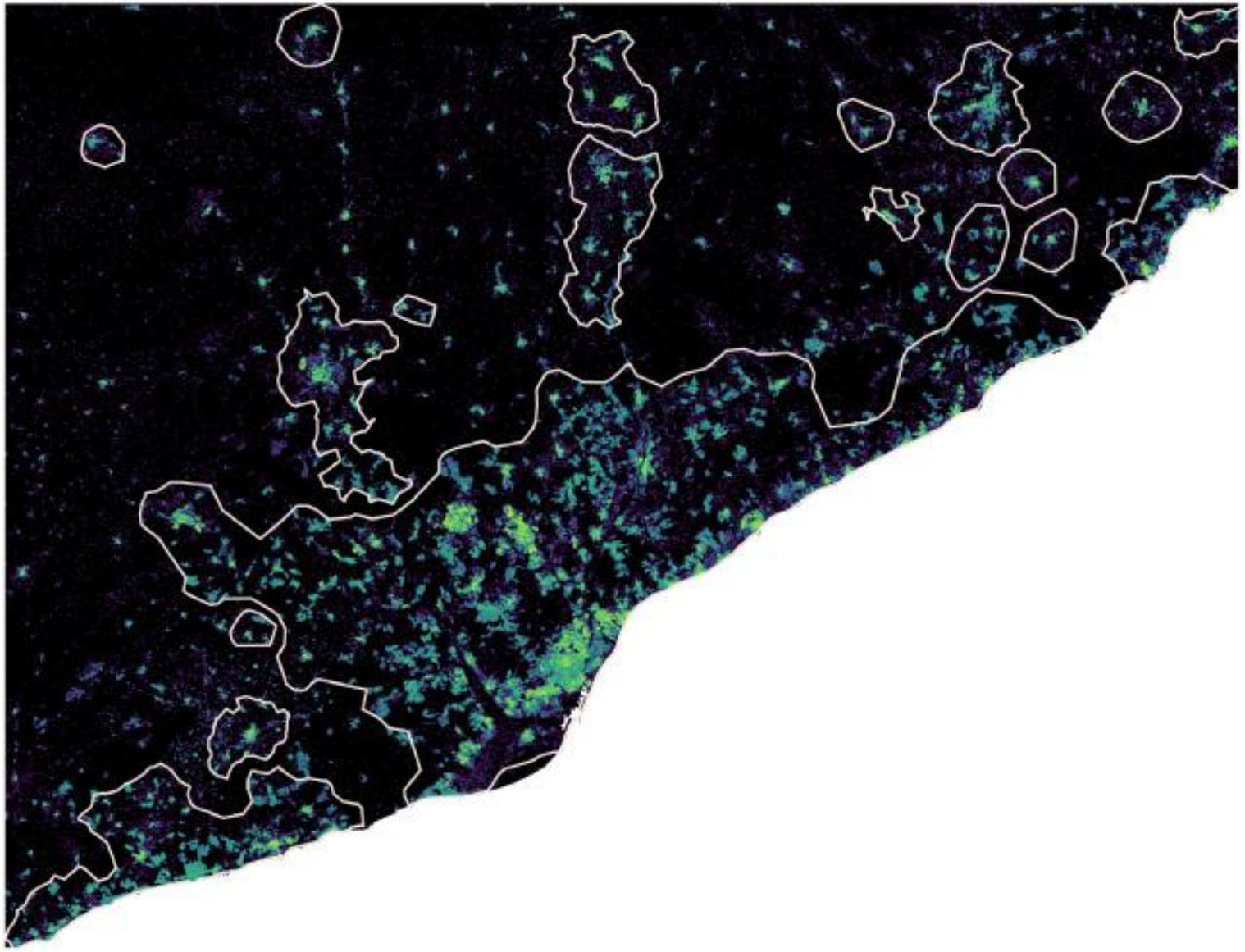
- Does not require specifying the number of clusters beforehand.
- Capable of finding clusters of any shape.

Cons:

- Results are highly dependent on the choice of `eps` and `minPts`.
- Struggles with datasets containing clusters of varying densities, as a single set of parameters may not accommodate all situations effectively.

DBSCAN

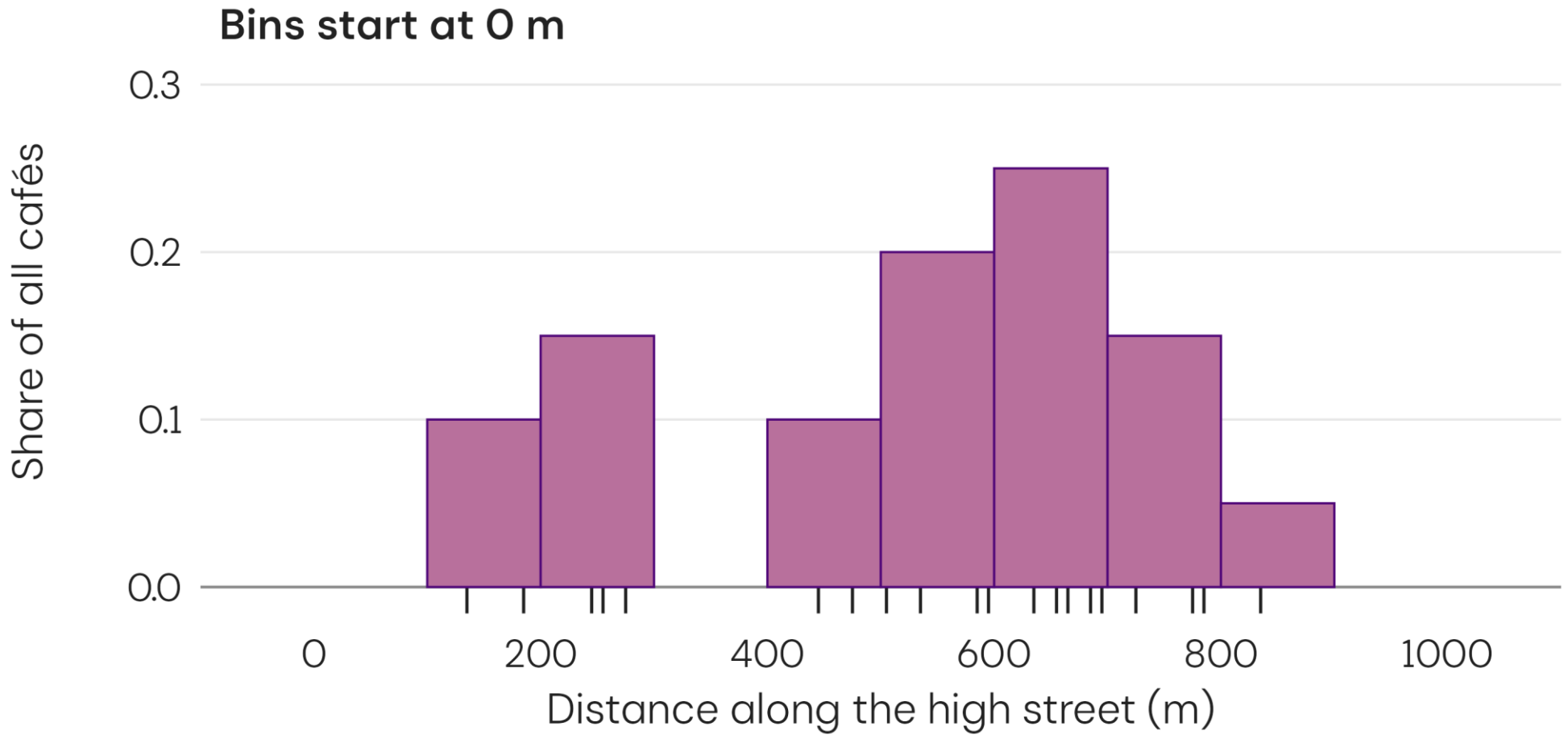
Arribas-Bel *et al.* 2021. Building(s and) cities: Delineating urban areas with a machine learning algorithm. *Journal of Urban Economics* 125: 103217



Kernel density estimation

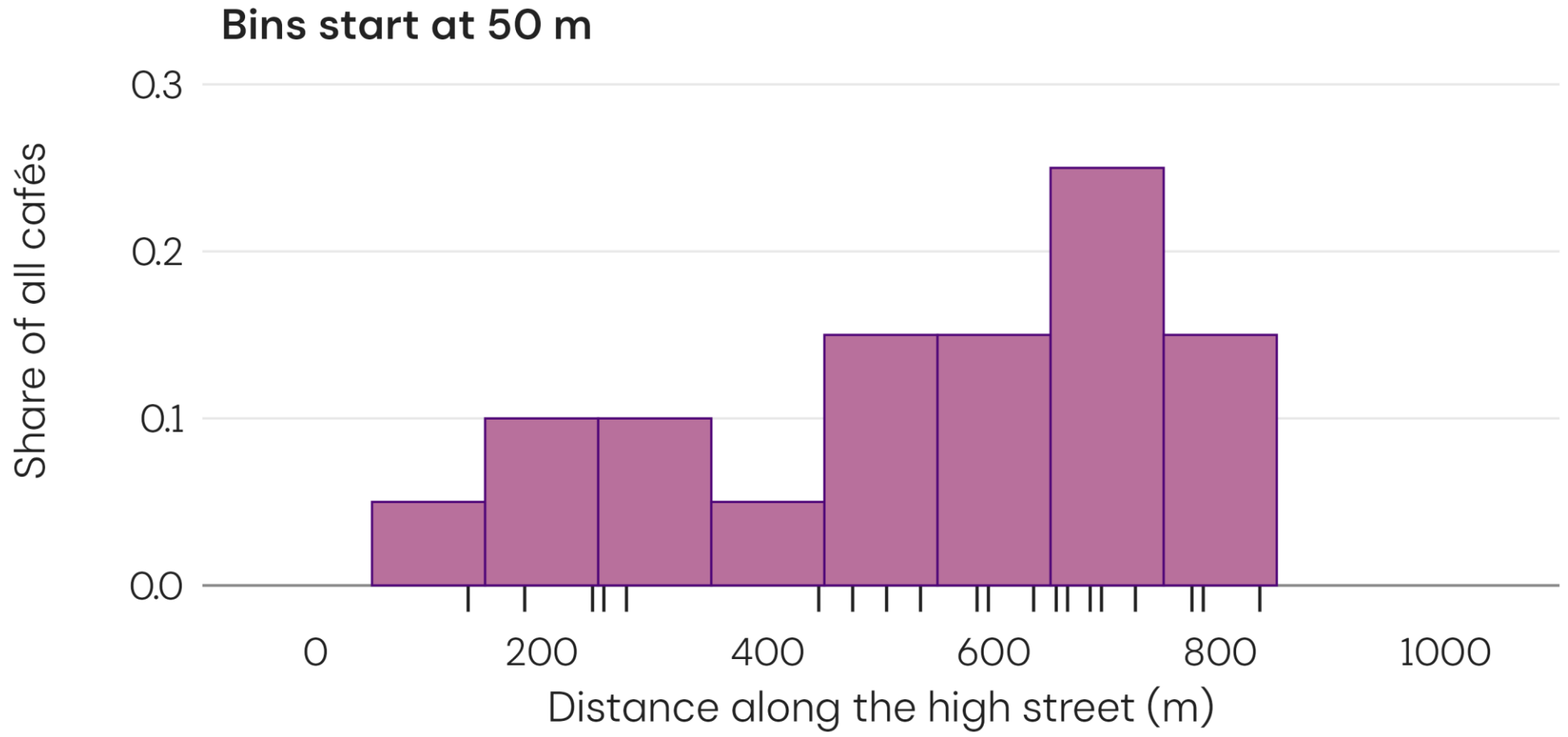
- Heatmap: Kernel density estimation (KDE) is a non-parametric way to estimate the probability density function of a random variable.
- It uses local information defined by windows (called kernels) to estimate densities of specified features at given locations.
- In essence it is a smoothing function where a continuous curve is created, based on a finite data sample.

A histogram estimates density



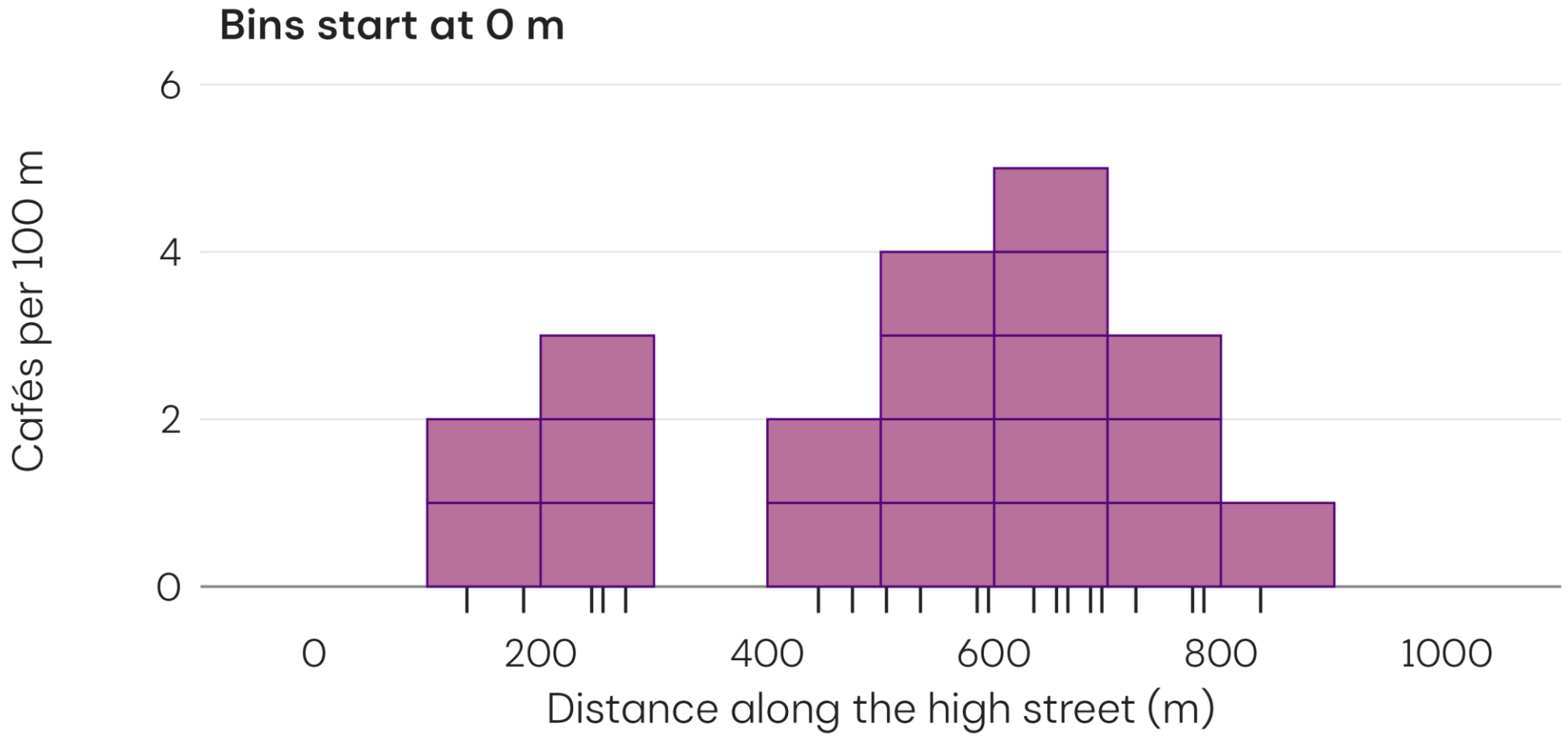
Simulated example: 20 cafés along a high street.

Same data, different bins



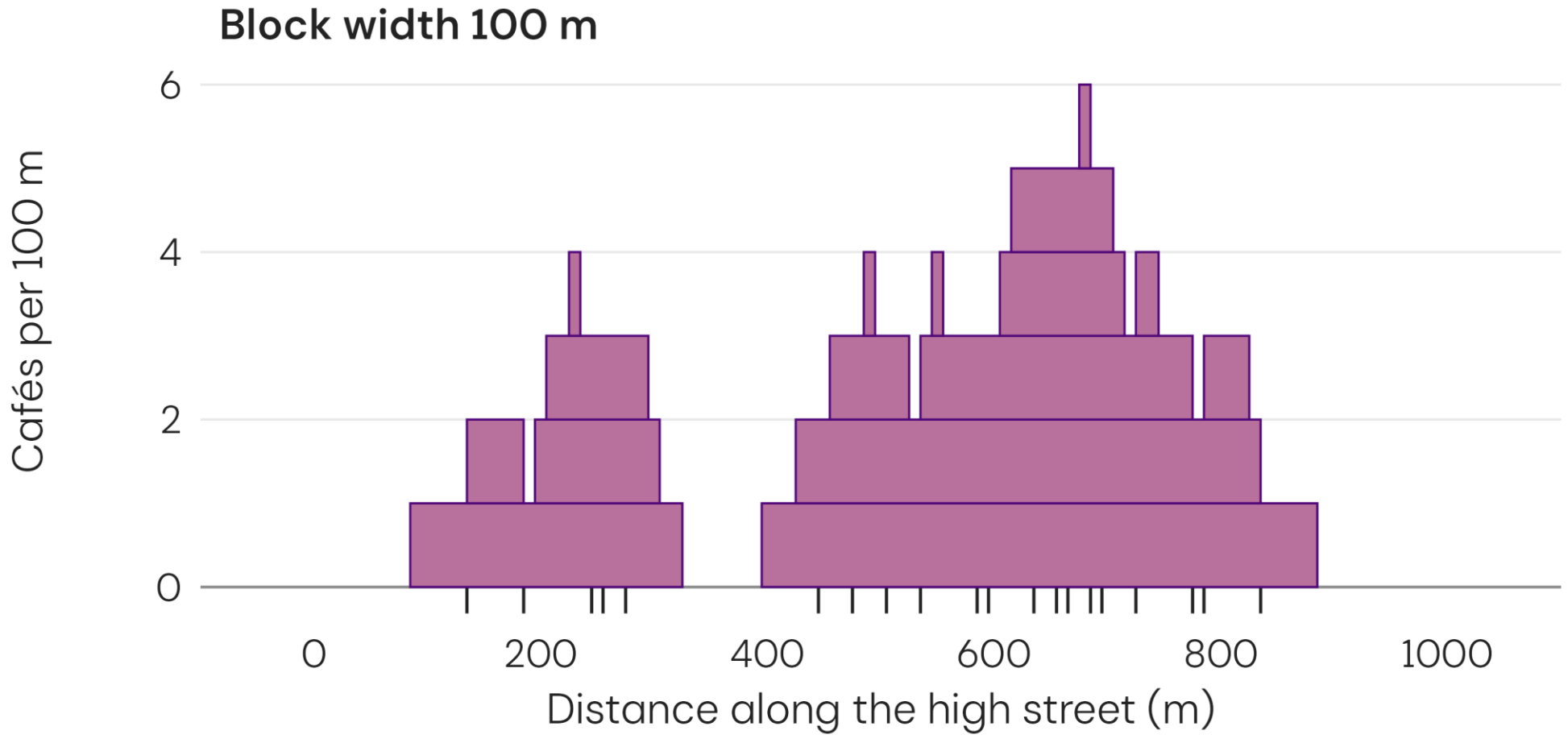
Simulated example: 20 cafés along a high street.

A histogram is a stack of blocks



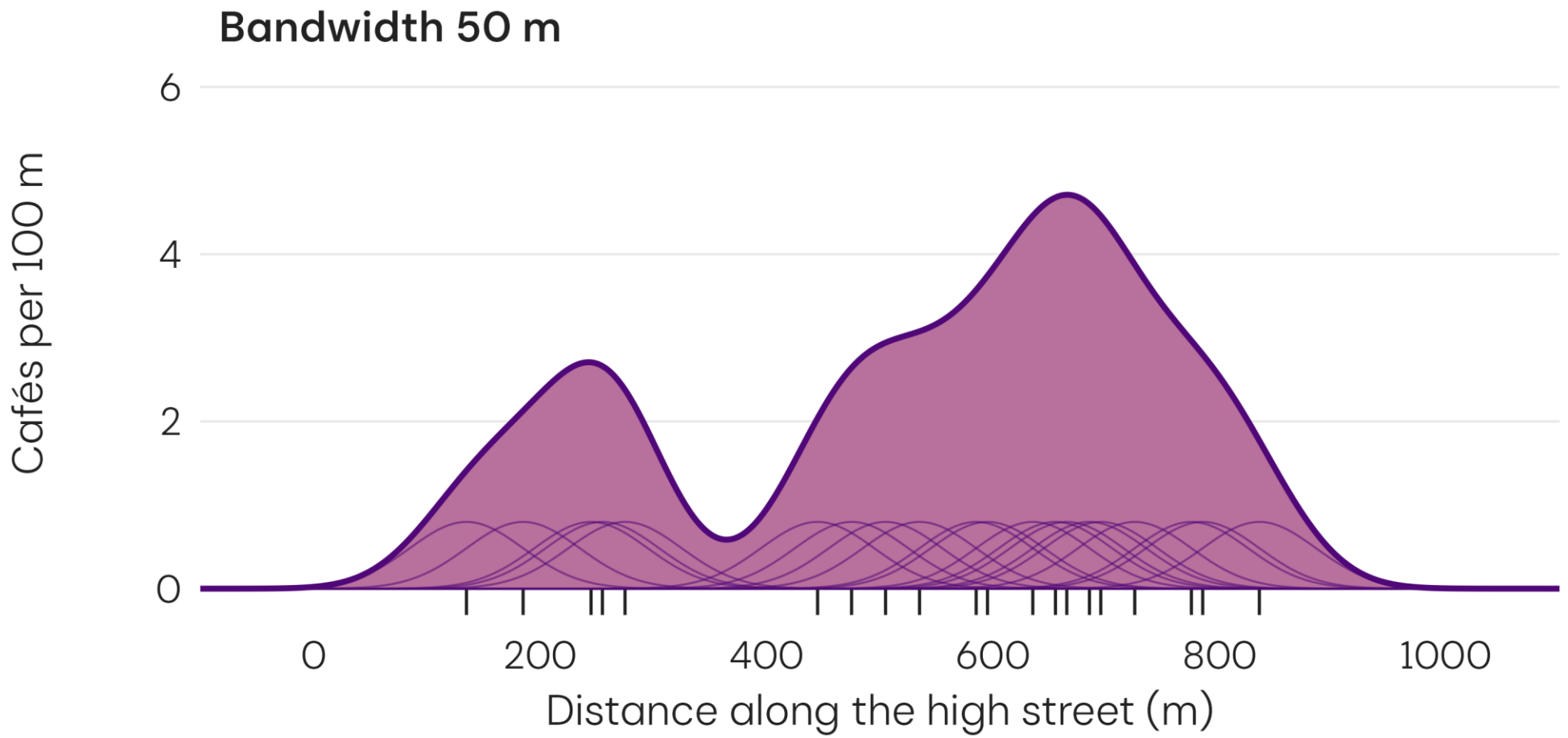
Simulated example: 20 cafés along a high street.

Stack the blocks on the points



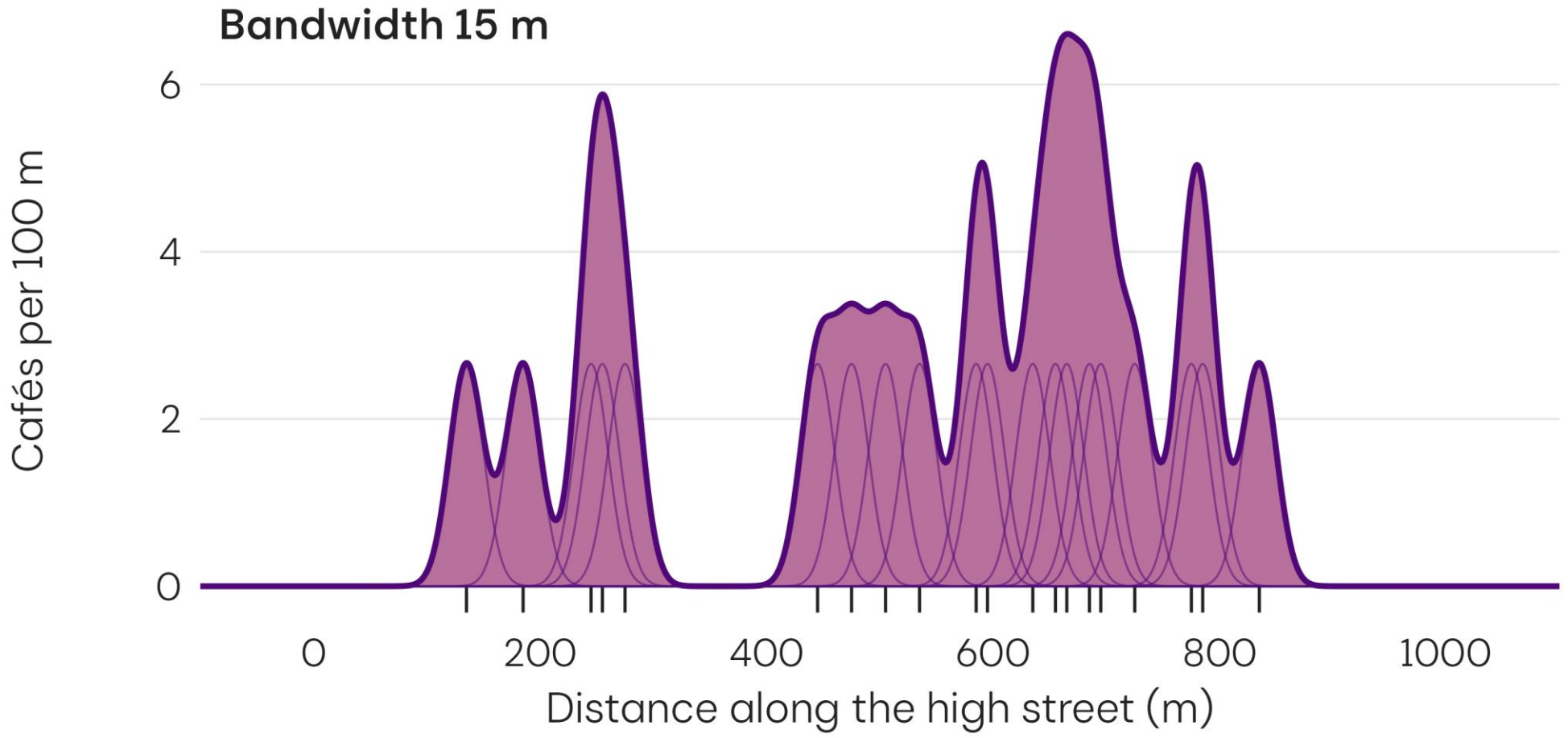
Simulated example: 20 cafés along a high street.

Replace each block with a smooth bump



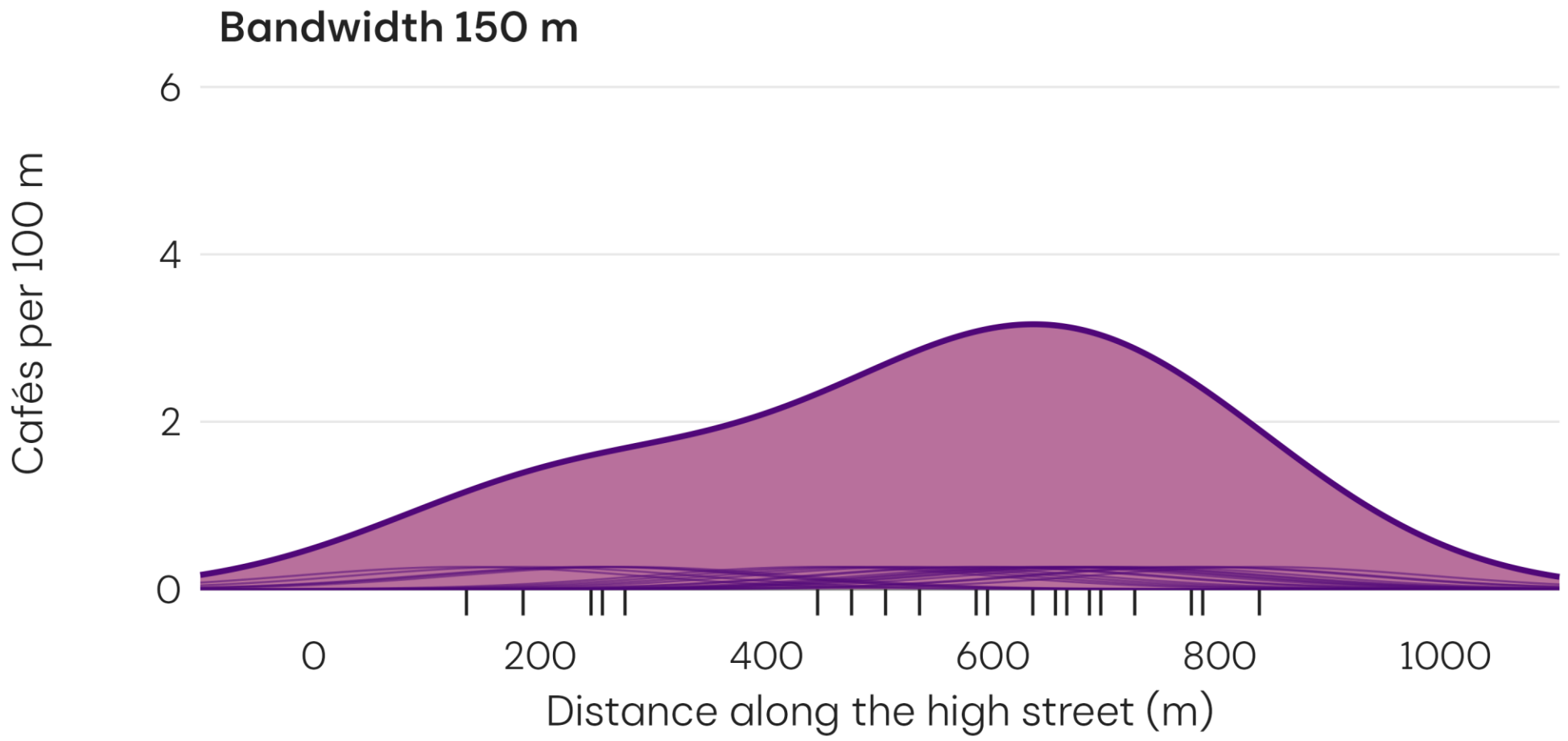
Simulated example: 20 cafés along a high street.

A narrower bump: a spikier curve



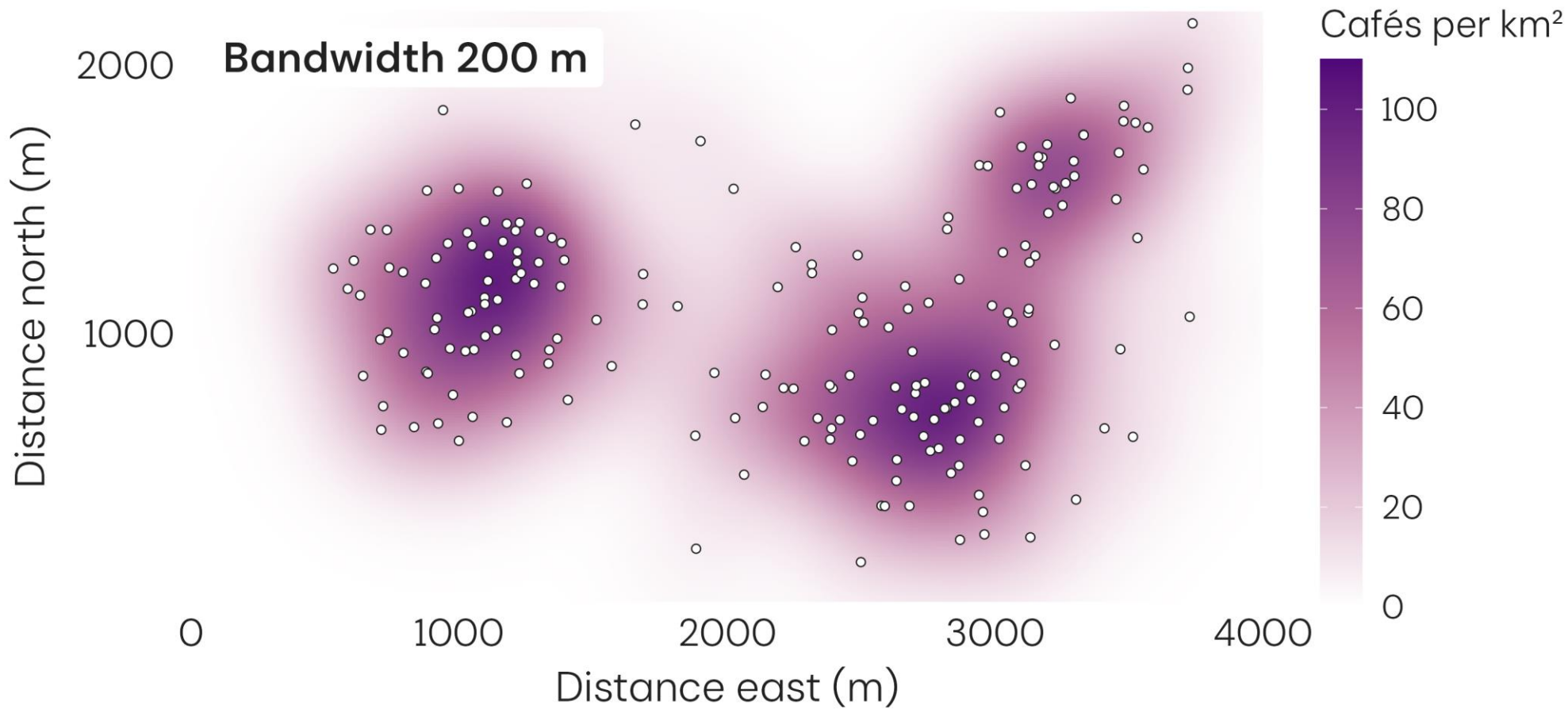
Simulated example: 20 cafés along a high street.

A wider bump: a smoother curve



Simulated example: 20 cafés along a high street.

The same idea in two dimensions



Simulated example: 200 cafés across a district.

Bandwidth: how wide each smooth bump is

- The kernel is the smooth bump placed on each point. The bandwidth is how wide that bump is.
- Small bandwidth: a spiky surface, with a peak on almost every point.
- Large bandwidth: a smooth surface that hides local detail and spills density into the surroundings.
- There is no single correct value: try several, compare the maps, and be able to explain your choice.

Kernel density estimation

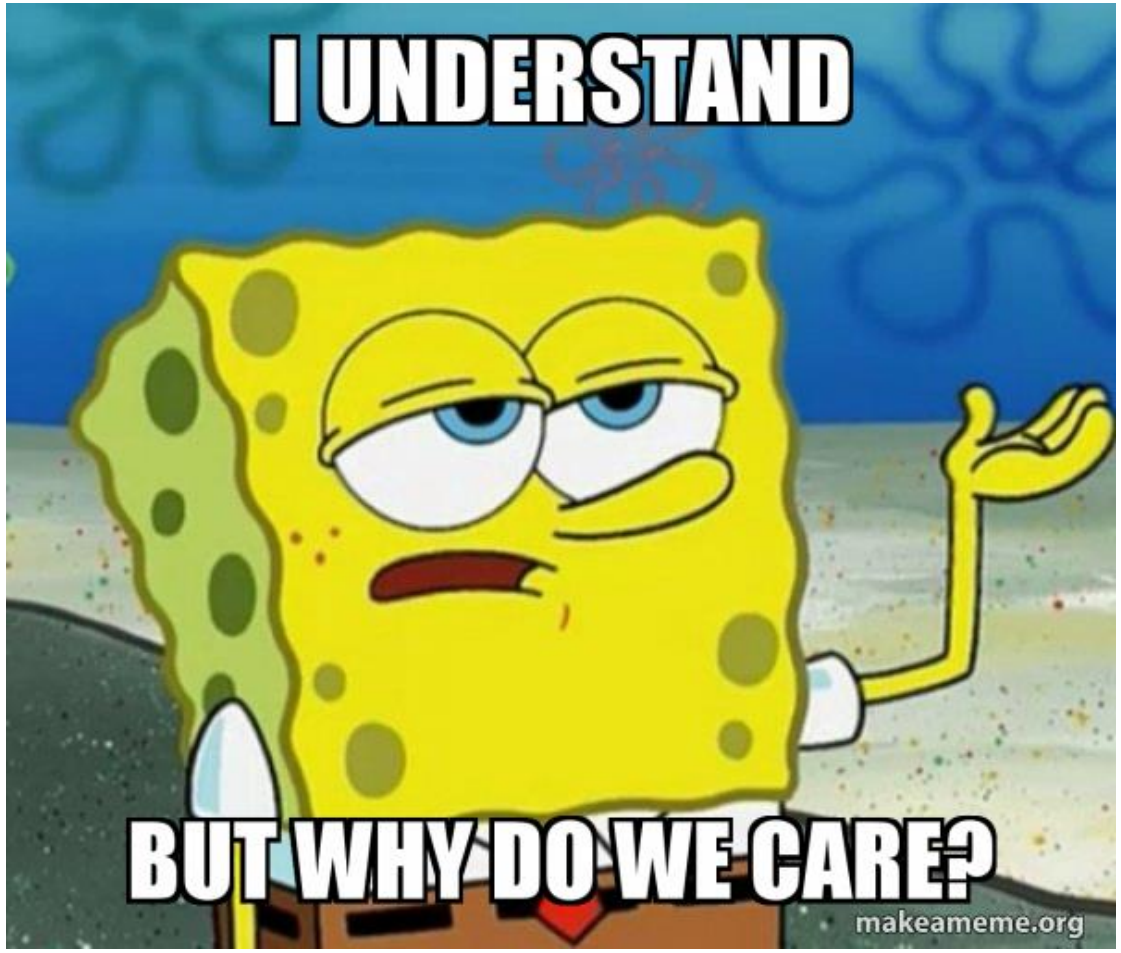
Pros:

- Produces a smooth surface that highlights hotspots without needing to define clusters beforehand.
- Simple to apply and widely used for visualising spatial concentration patterns.

Cons:

- Highly sensitive to the choice of bandwidth.
- Cannot handle areas with very different local densities well, because one global bandwidth may over-smooth or under-smooth (but adaptive bandwidths).

Kernel density estimation



Practical examples

Two examples

Point pattern analysis ‘in action’ in some actual research:

- Not all point data sets can be fully analysed with summary measures, e.g. analysis of GPS trajectory data (‘trajectory data’).
- Some problems do not per se look like a ‘point pattern analysis’ problem, but actually can be conceived as such, e.g. surname profiling.

GPS points along a day



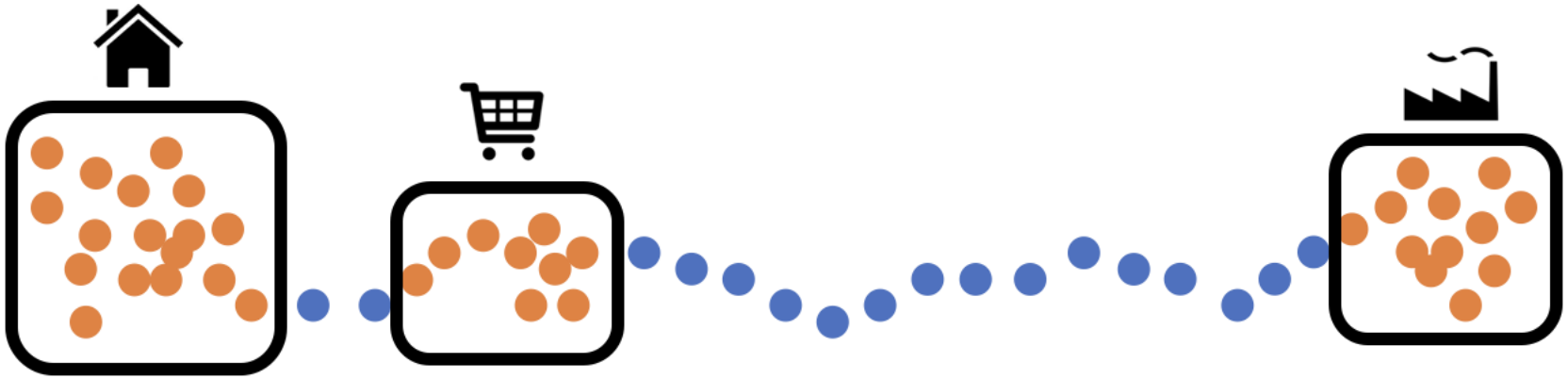
Van Dijk, J. T. 2018. Identifying activity-travel points from GPS-data with multiple moving windows. *Computers, Environment and Urban Systems* 70: 84-101

Stops are dense groups of points

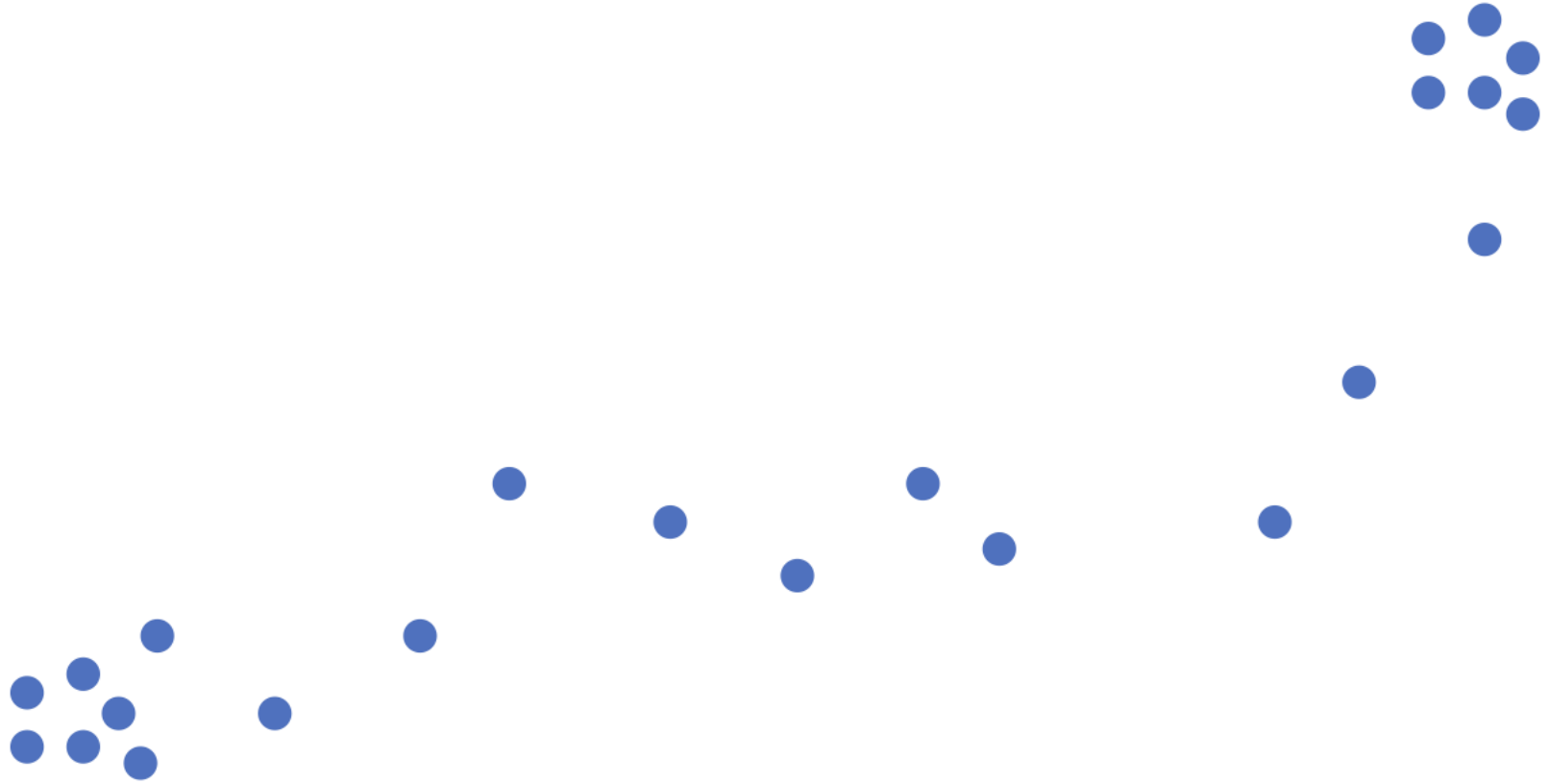


Van Dijk, J. T. 2018. Identifying activity-travel points from GPS-data with multiple moving windows. *Computers, Environment and Urban Systems* 70: 84-101

Stops become activities

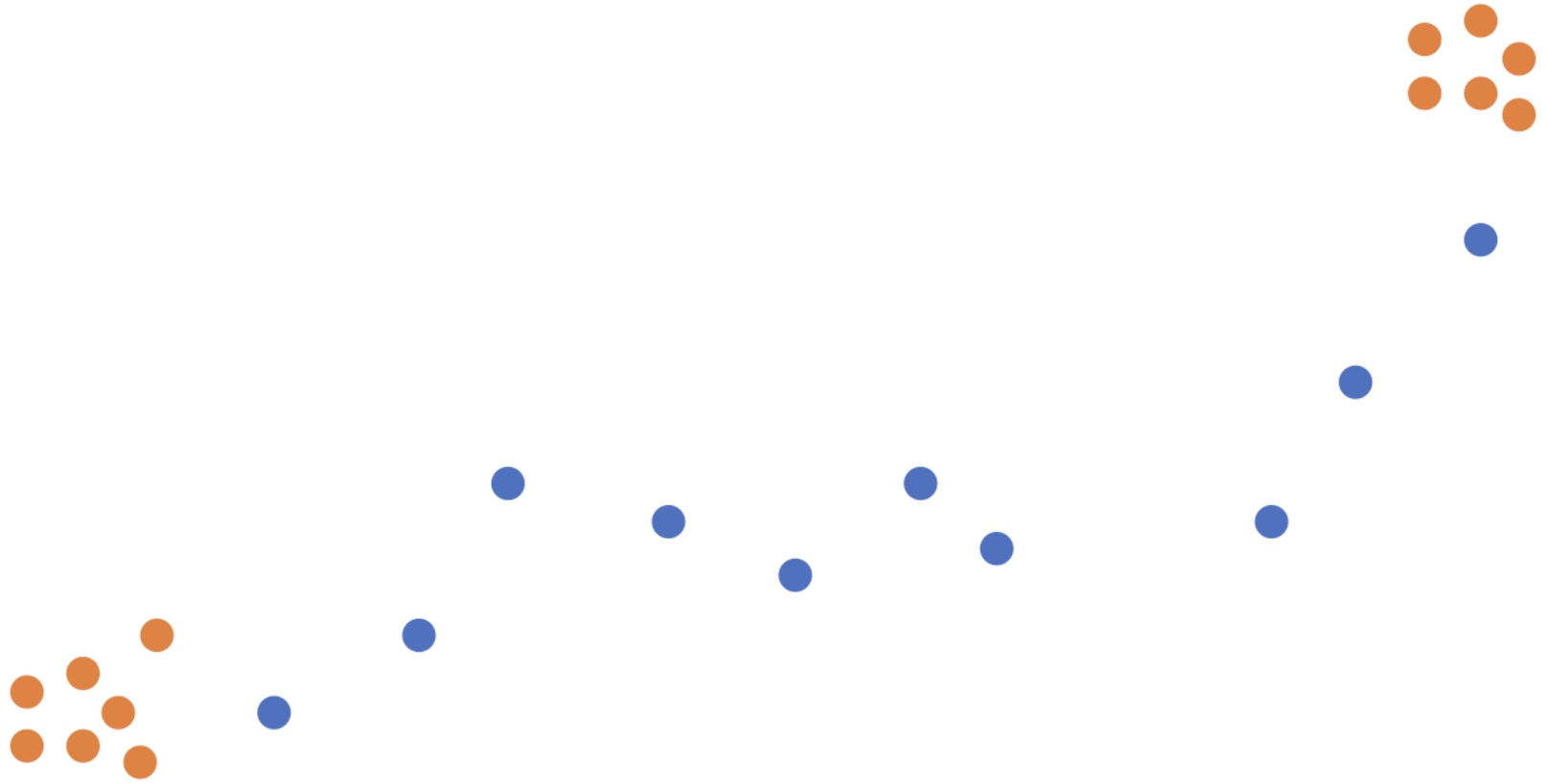


Reconstructing the route between stops



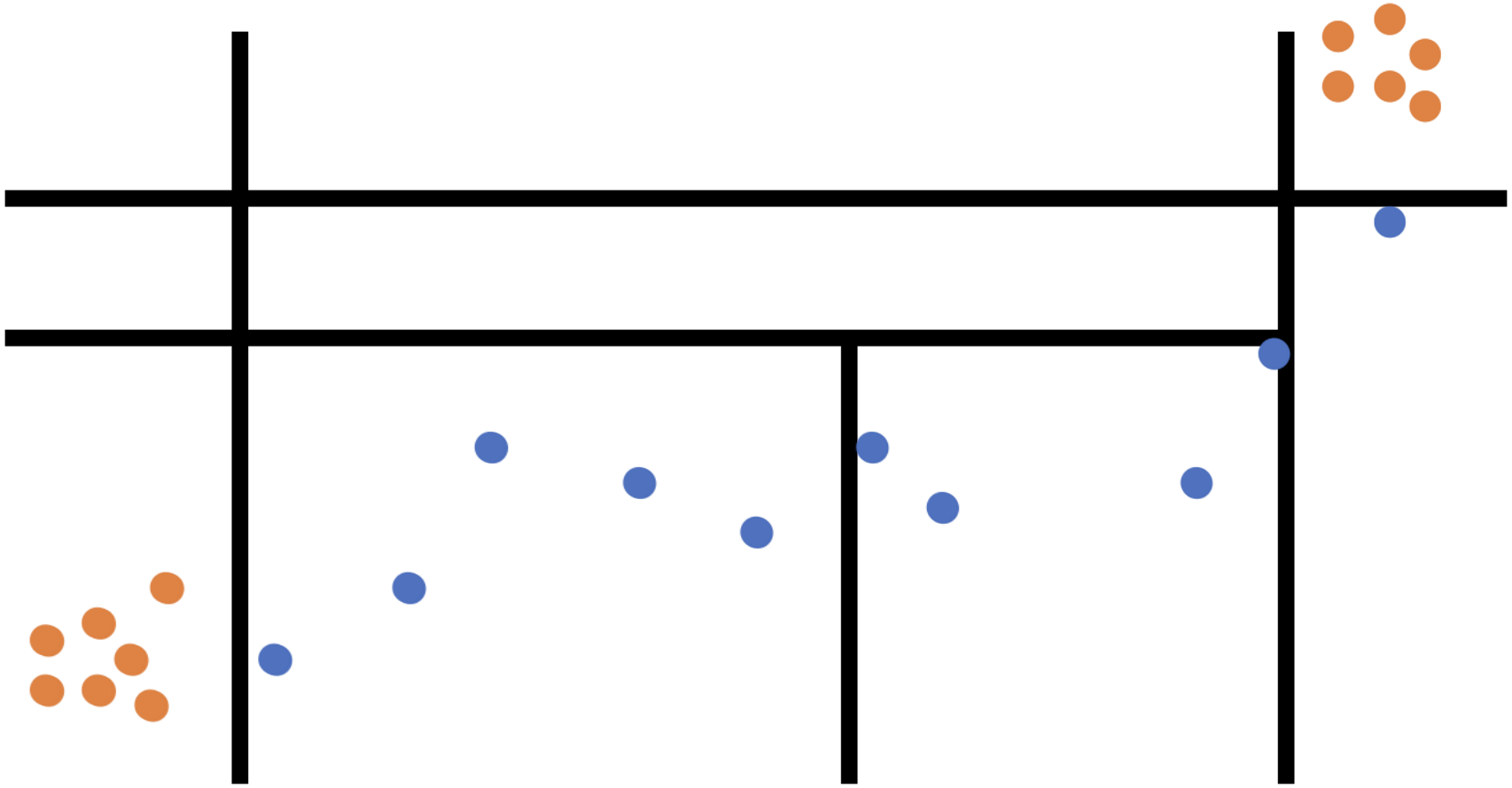
Van Dijk, J. T. & T. de Jong. 2017. Post-processing GPS tracks in reconstructing travelled routes in a GIS-environment: Network subset selection and attribute adjustment. *Annals of GIS* 23(3): 203-217

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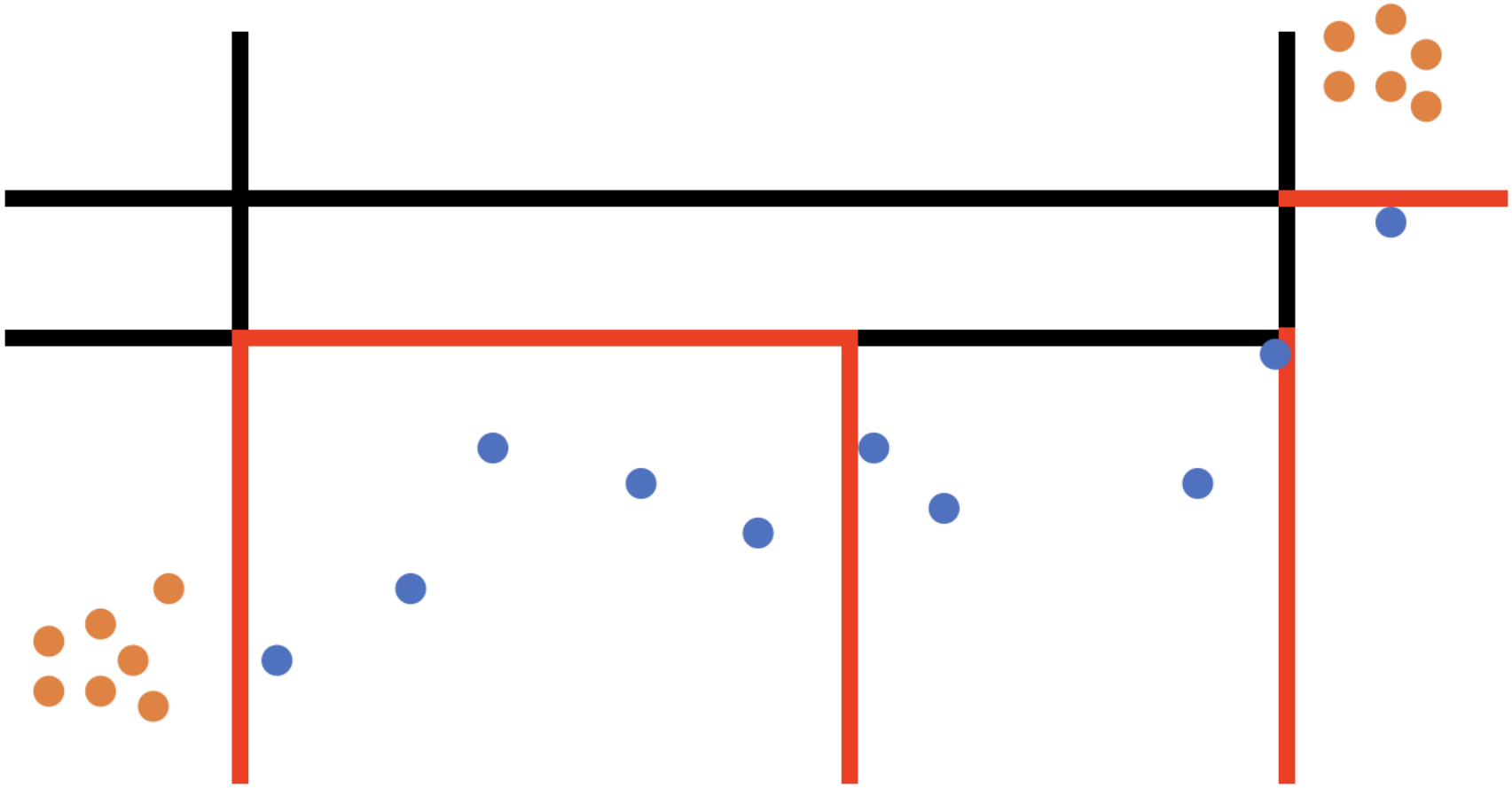
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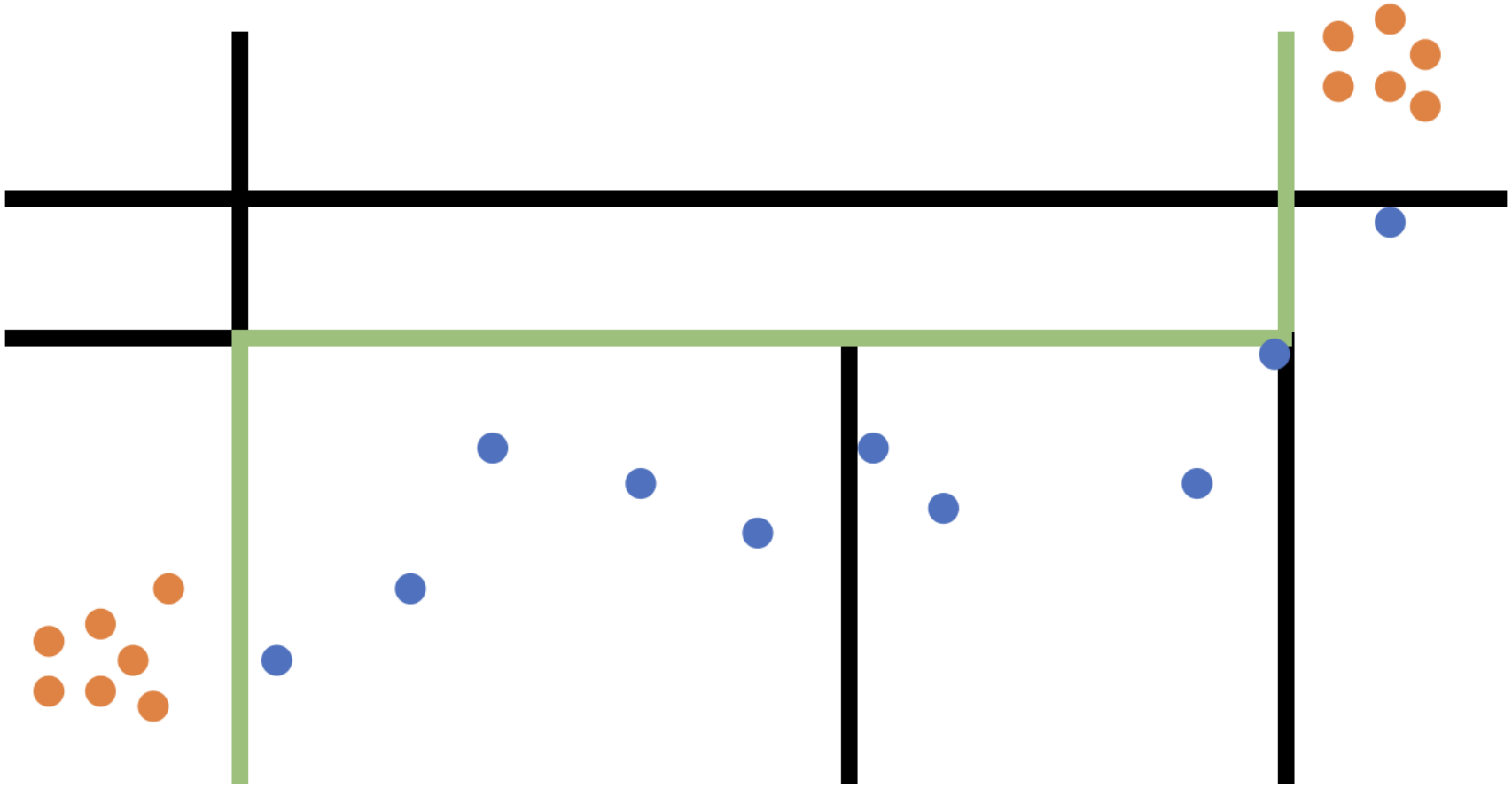
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Reconstructing the route between stops

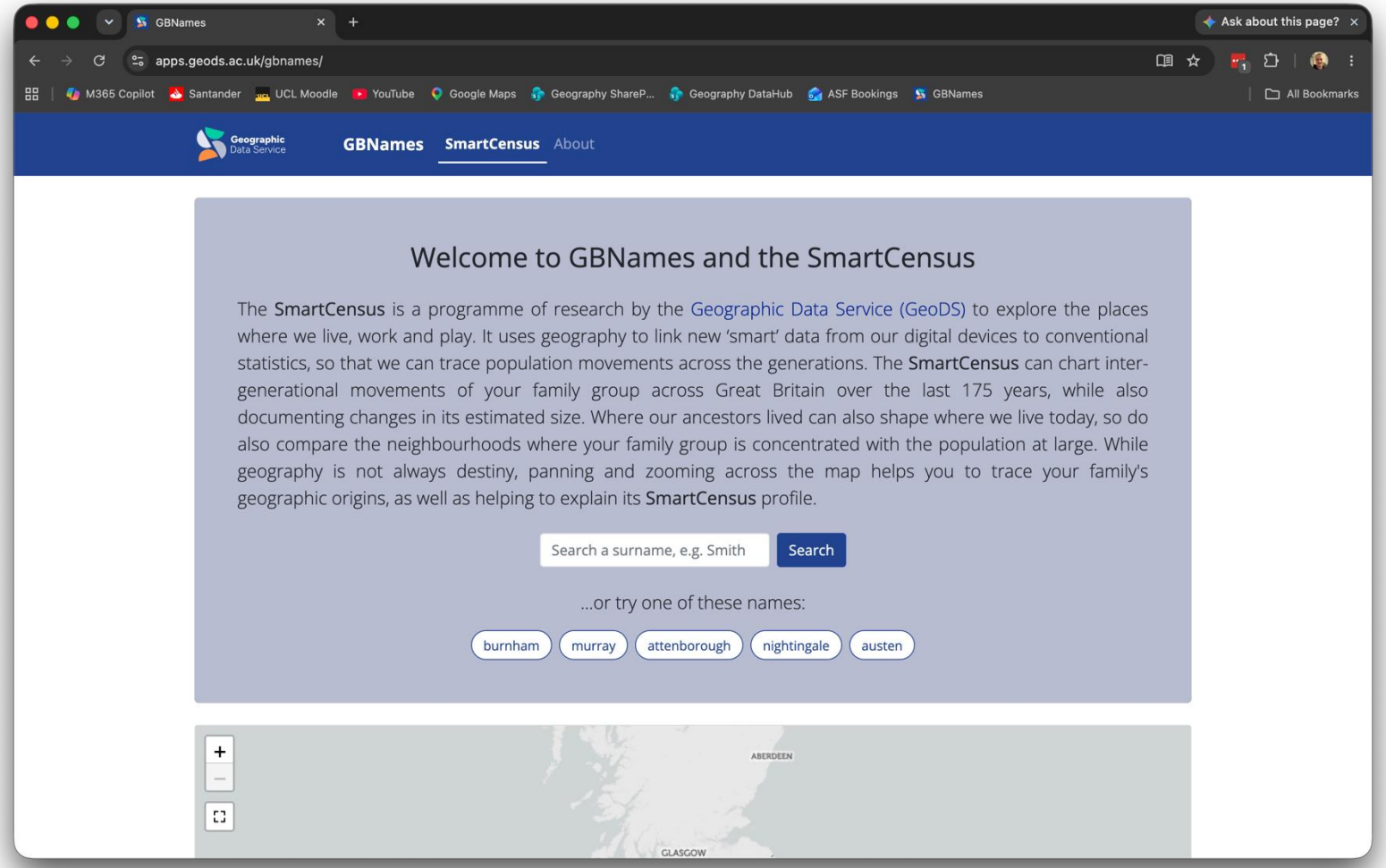


Van Dijk, J. T. & T. de Jong. 2017. Post-processing GPS tracks in reconstructing travelled routes in a GIS-environment: Network subset selection and attribute adjustment. *Annals of GIS* 23(3): 203-217

Surname profiling

- Personal names contain socio-demographic information.
- But also: many surnames contain spatial information at a variety of scales relating to the origins of many of their bearers.
- Data: Historic Census of Population 1851–1921, SmartCensus 1997–2026.
- Total of 400k surname distributions (one per surname for selected years, using Historic Parishes, unit postcodes and geo-coded addresses).

Surname profiling



Conclusion

- Point pattern analysis focuses on individual events in space, rather than aggregated counts in administrative units, where the boundaries can change the pattern (MAUP).
- KDE shows where events concentrate; DBSCAN shows which events form clusters and which are noise.
- Both need a choice you can justify: the bandwidth for KDE, `eps` and `minPts` for DBSCAN.
- Some datasets may not immediately look suited to point pattern analysis.

Questions

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